Eigen-Séta: an isogeny KEM whose trapdoor is a hidden-degree endomorphism

Specification, correctness, heuristic security analysis and a constant-time implementation

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Abstract

We specify Eigen-Séta v2, a key encapsulation mechanism built on supersingular isogenies in which the decryption trapdoor is a secret endomorphism of the public curve. The public key is a curve E_A together with an eigenbasis, on its 2^E-torsion, of the endomorphism $\tau = \varepsilon\omega\hat{\varepsilon}$, where ω is the degree-2 endomorphism of the curve with complex multiplication by $\mathbb{Z}[(1+\sqrt{-7})/2]$ and ε is an isogeny of *secret* degree f. A ciphertext is the codomain of a 3^F-isogeny together with the images of that eigenbasis under FESTA-style diagonal masks. The masks commute with the eigen-structure, so the holder of f decodes the ciphertext with a single $(2^E, 2^E)$-isogeny in dimension two (Kani's lemma), while to anyone else the ciphertext is a masked-torsion isogeny instance. We give the algorithms, prove correctness (including that the kernel used in decryption is always cyclic), derive parameters for target classical security $\lambda = 128, 192, 256$, and analyse key- and message-recovery attacks. The binding attack is guessing the mask ratio modulo $N \approx \sqrt{3^F}$, at cost about 2^λ classically and $2^{\lambda/2}$ Grover iterations; under NIST's definitions this is the same structure as AES key search, so we conjecture categories 1, 3 and 5. We also describe a Rust implementation with constant-time decapsulation. At the level-1 parameters (p of 1341 bits), public keys and ciphertexts are 1344 bytes and decapsulation takes about 14 s on one core. Instruction counts of the secret phase are identical across keys and across accepted and rejected ciphertexts. The security of the scheme is a conjecture supported by the analysis in this paper only; it has not been reviewed.

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1 Introduction

After the 2022 attacks on SIDH [CD23, MMP+23, Rob23], isogeny-based public-key encryption needs a trapdoor that does not reveal enough torsion information to let the attacker run those same attacks. Two families survive. In the first, torsion images are masked so that the attacker cannot recover them exactly, as in M-SIDH [FMP23] and FESTA [BMP23]; in the second, the scheme avoids torsion images altogether. FESTA decrypts by running the torsion attack itself with the help of a secret: the legitimate receiver knows a matrix that commutes with the masks.

Castryck and Vercauteren [CV23] observed that a torsion basis which is an eigenbasis of a *known* endomorphism of small degree makes masked-torsion instances easy: the masks act diagonally on the eigenvectors, so the endomorphism's action on the masked points is known exactly, and a lollipop endomorphism can be decoded with Kani's lemma. They presented this as an attack, and as a backdoor available to whoever chose the basis.

This work. Eigen-Séta turns that observation into a trapdoor. The public key is a curve E_A with an eigenbasis of an endomorphism $\tau \in \mathrm{End}(E_A)$ whose degree $2f^2$ is *secret*. The sender masks the images of the eigenbasis under a 3^F-isogeny φ with a random diagonal matrix. The receiver, who knows f, knows how $\theta = \varphi\tau\hat{\varphi}$ acts on the masked points and recovers φ from one $(2^E, 2^E)$-isogeny between products of elliptic curves. Without f, the ciphertext is a FESTA-type masked-torsion instance, and the public key is an isogeny instance of unknown degree from a known curve.

An earlier variant [Ano26] took $\tau = \varepsilon\omega\hat{\varepsilon}$ with ε of *public* degree 5^k. That variant is insecure: its decryption key is a function of the public degree alone (§6.2). Version 2, specified here, draws f from an interval of about 2^{277}, 2^{408} or 2^{535} values, depending on the level, and keeps it secret.

Contributions.

- A complete specification of the KEM (§3) with the Fujisaki–Okamoto transform with implicit rejection [HHK17].
- Proofs that decryption is correct (§4). These include a proposition that the kernel recovered during decryption is always cyclic, which settles a corner case reported for the earlier toy prototype.

- Parameter sets for $\lambda = 128, 192, 256$ (§5) and a heuristic analysis of key recovery, message recovery, quantum attacks, chosen-ciphertext attacks and decryption failures (§6). The concrete figures are summarised in Table 2.
- A Rust implementation with constant-time decapsulation, checked with an instruction-count probe and a statistical timing test. The probe found one compiler-introduced branch, which we fixed (§7).
- Measured sizes, timings and correctness rates at all parameter sets, and constant-time evidence at Toy16 and L1 (§8), each traceable to a committed record.

What this paper does not claim. Security is a conjecture backed by the attacks we considered; no third party has analysed the scheme. The implementation is research code. Its decapsulation is slow, mainly because the dimension-2 chain is computed in Mumford coordinates; a theta-coordinate chain [DMPR24] is future work. Key generation is not constant time.

Evidence conventions. Each measured number carries the path of the record it comes from, relative to the root of the repository (`crypto:`). Derived quantities are marked as such.

2 Preliminaries

Curves and isogenies. Let p be a prime and $\mathbb{F}_{p^2} = \mathbb{F}_p(i)$ with $i^2 = -1$ ($p \equiv 3 \bmod 4$). All curves are supersingular and defined over $\mathbb{F}_{p^2}$. Isogenies are separable. For an isogeny ϕ we write $\hat{\phi}$ for its dual, so that $\hat{\phi}\phi = [\deg \phi]$. For N coprime to p, the N-torsion $E[N] \cong (\mathbb{Z}/N)^2$ carries the Weil pairing e_N, and $e_N(\phi P, \phi Q) = e_N(P, Q)^{\deg \phi}$.

The base curve. Let E_D be the curve with j-invariant -3375. It has complex multiplication by $\mathcal{O}_K = \mathbb{Z}[(1 + \sqrt{-7})/2]$, and it is supersingular over $\mathbb{F}_p$ whenever -7 is not a square modulo p, i.e. $p \equiv 3, 5, 6 \pmod 7$. Its endomorphism ω of degree 2 satisfies $\omega^2 - t\omega + 2 = 0$ with $t = \pm 1$, so $(2\omega - t)^2 = -7$. We use the short Weierstrass model $y^2 = x^3 - 3jkx - 2jk^2$ with $j = -3375$ and $k = j - 1728$. Since $p + 1 = 2^{E+1}3^F c$, every curve in the isogeny class has $\#E(\mathbb{F}_{p^2}) = (p+1)^2$ and full rational 2^E- and 3^F-torsion.

Eigenbases. Let s_D be a square root of -7 modulo 2^{E+2} (it exists because $-7 \equiv 1 \bmod 8$), and set $\mu = (t + s_D)/2$ and $\nu = (t - s_D)/2$ modulo 2^E. These are the two roots of $x^2 - tx + 2$ modulo 2^E, and $\mu - \nu = s_D$ is odd. Hence ω has eigenvectors P_D, Q_D spanning $E_D[2^E]$ with $\omega P_D = \mu P_D$ and $\omega Q_D = \nu Q_D$.

Kani's lemma in the form used here. Let E be a curve, $\gamma \in \mathrm{End}(E)$ and $c' \in \mathbb{Z}$ with $\deg \gamma + c'^2 = 2^E$. Then

$$\Phi = \begin{pmatrix} c' & -\hat{\gamma} \\ \gamma & c' \end{pmatrix} : E^2 \to E^2$$

satisfies $\hat{\Phi}\Phi = [2^E]$. It is therefore a $(2^E, 2^E)$-isogeny between principally polarised abelian surfaces, and its kernel is $\{(c'R, -\gamma R) : R \in E[2^E]\}$. Given $\gamma(R)$ for a basis of $E[2^E]$, the isogeny Φ can be computed as a chain of E $(2,2)$-isogenies [Kan97, CD23, Rob23], and it can then be evaluated at points outside the 2^E-torsion. In particular $\Phi(X, 0) = (c'X, \gamma X)$ reveals γ on any subgroup.

$(2,2)$-chains. A chain from $E \times E$ can pass through products of elliptic curves (product steps), glue into the Jacobian of a genus-2 curve, continue with Richelot isogenies, and split back into a product. We use the gluing, Richelot and splitting formulas of Castryck and Decru [CD23] in Mumford coordinates. The order in which these step types occur is the *schedule* of the chain.

FESTA masking. In FESTA [BMP23], the images of a torsion basis under a secret isogeny are published multiplied by a secret matrix. We use diagonal masks: $(a\varphi P, b\varphi Q)$ with odd a, b modulo 2^E. The Weil pairing reveals ab. The ratio b/a stays hidden.

3 The scheme

3.1 Parameters

A parameter set consists of the following data.

- Exponents E and F and a cofactor c such that $p = 2^{E+1}3^F c - 1$ is prime and $p \equiv 3, 5, 6 \pmod 7$.
- $d = 3^F$, the sender's isogeny degree.
- A lower bound f_{lo} for the secret degree, which ranges over $[f_{\text{lo}}, 2f_{\text{lo}})$.
- The curve E_D, its endomorphism ω with trace t, the values s_D, μ, ν and the eigenbasis (P_D, Q_D) of §2.
- Domain-separated instances of SHAKE256: G (masks), H_P (public-key hash), H_K (shared secret) and H_R (rejection key).

The concrete sets are in §5.

Canonical bases. $\mathsf{Basis}_3(A)$ deterministically computes a basis (P_3, Q_3) of $E[3^F]$ on the Montgomery curve $E : y^2 = x^3 + Ax^2 + x$. It tries $x = 1 + i, 2 + i, \ldots$, lifts each to a point, multiplies by $(p+1)/d$, and keeps the first two points whose 3^{F-1}-multiples generate $E[3]$. It depends only on A, so sender and receiver agree on it.

3.2 Key generation

The degree-f factor ε exists by Kani's lemma applied to α, since $\deg \alpha + f^2 = f \cdot 2^E$ and $\gcd(f, g) = 1$ (see [Rob23, §2]). The requirement $(y_0, y_1) \neq (0, 0)$ excludes $\alpha \in \mathbb{Z}[\omega]$, for which E_A would lie in the small orbit of E_D under the class group of $\mathcal{O}_K$ (§6.7). The endomorphism the public key is built around is

$$\tau = \varepsilon\omega\hat{\varepsilon} \in \mathrm{End}(E_A), \qquad \deg \tau = 2f^2, \quad \mathrm{Tr}\, \tau = tf,$$

and since $\tau\varepsilon = f\,\varepsilon\omega$, the points P_A, Q_A are eigenvectors of τ on $E_A[2^E]$ with eigenvalues $f\mu$ and $f\nu$. The scalars λ_1, λ_2 make the Weil pairing $e(P_A, Q_A) = e(P_D, Q_D)^{\lambda_1\lambda_2 f}$ uniform among values of the same order, so it carries no information on f.

3.3 Encapsulation

3.4 Decapsulation

Write $S = aP'$ and $T = bQ'$, and $\theta = \varphi\tau\hat{\varphi} \in \mathrm{End}(E_C)$. Section 4 shows that $\psi = 2\theta - tdf$ has degree $7d^2f^2$ and trace 0, and acts on S and T by $\pm ds_0$. Decapsulation applies Kani's lemma to $\gamma = \psi + c$, whose degree is $7d^2f^2 + c^2 = 2^E - c'^2$.

All steps after line 2 run in constant time with respect to sk and s (§7). Lines 1–2 depend only on public data.

Algorithm 1 KeyGen()

repeat ▷ secret degree and Kani scalars
  $f \leftarrow$ uniform odd integer in $[f_{\mathrm{lo}}, 2f_{\mathrm{lo}})$
  reject if $3 \mid f$, $7 \mid f$ or $2f \equiv 2^E \pmod d$
  $m \leftarrow 2^E - 7d^2f^2$; find c, c' with $c^2 + c'^2 = m$, c even, or reject
  $s_0 \leftarrow fs_D \bmod 2^E$
  choose signs of c, c' with $\max(v_2(c' - \lambda_S), v_2(c' + \lambda_S)) = 2$, where $\lambda_S = ds_0 + c$, or reject
until accepted
$g \leftarrow 2^E - f$
repeat ▷ endomorphism of norm fg that uses Frobenius
  pick $(y_0, y_1) \neq (0, 0)$ with $T \leftarrow fg - p\,N(y_0 + y_1\omega) > 0$ prime, $T \equiv 1, 2, 4 \pmod 7$
  solve $x^2 + 7y'^2 = T$ (Cornacchia); $x_0 \leftarrow x - ty'$, $x_1 \leftarrow 2y'$
until found
$\alpha \leftarrow x_0 + x_1\omega + (y_0 + y_1\omega)\pi$ ▷ π the p-Frobenius; $\operatorname{nrd}\alpha = fg$
compute the $(2^E, 2^E)$-isogeny with kernel $\{([f]R, \alpha R) : R \in E_D[2^E]\}$; reject unless its first step is a gluing
from its codomain $E_1 \times E_2$, pick the factor E_A whose map $\varepsilon : E_D \to E_A$ has degree f (test $e(\varepsilon X, \varepsilon Y) = e(X, Y)^f$ on $E_D[3^F]$)
$\lambda_1, \lambda_2 \leftarrow$ uniform odd residues modulo 2^E
$P_A \leftarrow \lambda_1\varepsilon(P_D)$, $Q_A \leftarrow \lambda_2\varepsilon(Q_D)$; bring E_A to Montgomery form $y^2 = x^3 + Ax^2 + x$
$\lambda_T \leftarrow c - ds_0$; reject if λ_S, λ_T or $c' \equiv -1 \pmod{2^E}$, or c or $c' \equiv -1 \pmod d$
$pk \leftarrow (A, x(P_A), x(Q_A), x(P_A - Q_A))$; $z \leftarrow$ 32 random bytes
$sk \leftarrow (\lambda_S, \lambda_T, c' \bmod 2^E, c \bmod d, c' \bmod d, z, pk)$
return (pk, sk) after a self-test encapsulation and decapsulation

3.5 The transform

Enc is a deterministic public-key encryption of the message $s \in \mathbb{Z}/d$, derandomised by deriving the masks from $G(s, H_P(pk))$. The KEM is the $U^{\not\perp}$ transform with implicit rejection of Hofheinz, Hövelmanns and Kiltz [HHK17] applied to it. The re-encryption check compares the entire ciphertext, including the curve coefficient.

4 Correctness

Throughout, (pk, sk) is honestly generated, $ct = \mathsf{Enc}(pk, s)$, and the notation of Algorithms 1–3 is in force. Write $\theta = \varphi\tau\hat{\varphi} \in \mathrm{End}(E_C)$ and $\psi = 2\theta - tdf$.

Lemma 1 (the lollipop endomorphism)**.** $\operatorname{Tr}\psi = 0$ *and* $\deg\psi = 7d^2f^2$*. Moreover* $\psi S = ds_0 S$ *and* $\psi T = -ds_0 T$*, where* $s_0 = fs_D \bmod 2^E$*.*

Proof. $\operatorname{Tr}\theta = \varphi(\tau + \hat{\tau})\hat{\varphi} = \varphi[tf]\hat{\varphi} = tdf$, and $\deg\theta = d^2\deg\tau = 2d^2f^2$. Hence $\operatorname{Tr}\psi = 2\operatorname{Tr}\theta - 2tdf = 0$ and

$$\deg\psi = (2\theta - tdf)(2\hat{\theta} - tdf) = 4\deg\theta - 2tdf\operatorname{Tr}\theta + t^2d^2f^2 = 8d^2f^2 - 2d^2f^2 + d^2f^2 = 7d^2f^2.$$

For the eigenvalues, $\hat{\varphi}S = a\hat{\varphi}\varphi P_A = adP_A$, and $\tau P_A = f\mu P_A$, so $\theta S = adf\mu\,\varphi P_A = df\mu S$. Likewise $\theta T = df\nu T$. Then $\psi S = df(2\mu - t)S = df s_D S$, and $\psi T = df(2\nu - t)T = -df s_D T$. □

The masks a, b drop out because S and T are multiples of eigenvectors. This is where the trapdoor is used: only f (through s_0) and the Kani scalars are needed, never φ or the masks.

Proposition 2 (the decryption isogeny)**.** *Let* $\gamma = \psi + c$*. Then* $\deg\gamma + c'^2 = 2^E$*, and the subgroup* $\langle (c'S, -\lambda_S S), (c'T, -\lambda_T T)\rangle$ *of* E_C^2 *is the kernel of the* $(2^E, 2^E)$*-isogeny* $\Phi = \left(\begin{smallmatrix} c' & -\hat{\gamma} \\ \gamma & c' \end{smallmatrix}\right)$.

Algorithm 2 $\mathsf{Encaps}(pk)$ and the deterministic core $\mathsf{Enc}(pk, s)$

$s \leftarrow$ uniform in $\mathbb{Z}/d$; $ct \leftarrow \mathsf{Enc}(pk, s)$; **return** $(ct,\ H_K(s, ct))$

function $\textsc{Enc}(pk = (A, x_P, x_Q, x_{P-Q}), s)$
  $(P_3, Q_3) \leftarrow \mathsf{Basis}_3(A)$; $K \leftarrow P_3 + sQ_3$ ▷ 3-point ladder
  $\varphi : E_A \to E_C = E_A/\langle K \rangle$ ▷ chain of F 3-isogenies, fixed strategy
  push x_P, x_Q, x_{P-Q} through φ: $P' = \varphi(P_A)$, $Q' = \varphi(Q_A)$
  $(a, b) \leftarrow G(s, H_P(pk))$, odd residues modulo 2^E
  return $(A_C,\ x(aP'),\ x(bQ'),\ x(aP' - bQ'))$
end function

Algorithm 3 $\mathsf{Decaps}(sk, ct)$

parse $ct = (A_C, x_S, x_T, x_{S-T})$; lift S, T on E_C with $x(S - T) = x_{S-T}$; on failure **return** $H_R(z, ct)$ ▷ public
$(X_1, X_2) \leftarrow \mathsf{Basis}_3(A_C)$; $(P_3, Q_3) \leftarrow \mathsf{Basis}_3(A)$ ▷ public
$\Phi \leftarrow$ the $(2^E, 2^E)$-isogeny of E_C^2 with kernel $\langle (c'S, -\lambda_S S), (c'T, -\lambda_T T) \rangle$, fixed schedule
evaluate Φ at $(X_1, 0), (0, X_1), (X_2, 0)$; identify both codomain factors with E_C among the 8 sign and order choices by comparing with $c'X_i$; read off $\gamma(X_1), \gamma(X_2)$
$\psi(X_i) \leftarrow \gamma(X_i) - cX_i$ for $i = 1, 2$
$G \leftarrow$ whichever of $\psi(X_1), \psi(X_2)$ has order d; $H \leftarrow$ the other; find k with $H = kG$ ▷ pairings, DLP in μ_d
$\hat{\varphi} \leftarrow E_C/\langle X_h - kX_g \rangle \to E_A$; $K_g \leftarrow$ the image of X_1 or X_2 that has order d
$s \leftarrow \log(e(P_3, K_g))/\log(e(K_g, Q_3))$ in μ_d
$ct' \leftarrow \mathsf{Enc}(pk, s)$
return $H_K(s, ct)$ if $ct' = ct$, else $H_R(z, ct)$ ▷ constant-time selection

Proof. $\deg \gamma = \deg \psi + c \operatorname{Tr} \psi + c^2 = 7d^2 f^2 + c^2 = 2^E - c'^2$ by the choice of c, c'. By Lemma 1, $\gamma S = (ds_0 + c)S = \lambda_S S$ and $\gamma T = (c - ds_0)T = \lambda_T T$. The points S, T form a basis of $E_C[2^E]$, because φ has odd degree and a, b are odd. So the two generators are $(c'R, -\gamma R)$ for $R = S, T$, and the kernel statement is Kani's lemma (§2). □

From $\Phi(X, 0) = (c'X, \gamma X)$, decapsulation obtains γ, and hence $\psi = \gamma - c$, on $E_C[d]$. Restricted to $E_C[d]$ we have $\psi = 2\varphi\tau\hat{\varphi}$, since $d \mid tdf$.

Proposition 3 (the recovered kernel is always cyclic)**.** *$\psi(E_C[d])$ is cyclic of order d, and the kernel of ψ restricted to $E_C[d]$ is exactly* $\ker \hat{\varphi}$.

Proof. We have $\hat{\varphi}(E_C[d]) = \ker \varphi = \langle K \rangle$, which is cyclic of order d. The endomorphism τ has degree $2f^2$, which is prime to 3, so $\tau\langle K \rangle$ is again cyclic of order d. The map φ is injective on $\tau\langle K \rangle$ if and only if $\tau\langle K \rangle \cap \langle K \rangle = 0$, i.e. if and only if the two subgroups have different subgroups of order 3. That fails only if the line $L = \langle 3^{F-1}K \rangle \subset E_A[3]$ is an eigenline of τ modulo 3. The characteristic polynomial of τ is $x^2 - tfx + 2f^2$, whose discriminant is $f^2(t^2 - 8) = -7f^2 \equiv 2f^2 \pmod 3$. Since $3 \nmid f$, this is a non-square modulo 3. So τ has no eigenline in $E_A[3]$, and φ maps $\tau\langle K \rangle$ isomorphically onto a cyclic group of order d. Thus $\psi(E_C[d]) = 2\varphi\tau\langle K \rangle$ has order d, its kernel has order $d^2/d = d$, and that kernel contains $\ker \hat{\varphi}$, which also has order d. □

Remark 4. The design note [Ano26] reported a "non-cyclic ψ'" corner case in one of eight trials of its toy prototype, a variant with a different ψ' and very small $d = 3^5$. Proposition 3 shows that the case cannot occur in version 2. The argument needs only that -7 is a non-square modulo 3 and $3 \nmid \deg \tau$. The cause of the toy observation was not isolated.

Proposition 5 (recovery of s)**.** *Let G be a generator of $\psi(E_C[d])$ and $H = kG$ the image of the other basis point. Then $\ker\hat{\varphi} = \langle X_h - kX_g\rangle$. If K_g is the image under $\hat{\varphi}$ of a basis point and has order d, then $K_g = u(P_3 + sQ_3)$ for a unit u, and $s = \log e(P_3, K_g)/\log e(K_g, Q_3)$ in μ_d.*

Proof. $\psi(X_h - kX_g) = H - kG = 0$, and $X_h - kX_g$ has order d, so by Proposition 3 it generates $\ker\hat{\varphi}$. The image of $E_C[d]$ under $\hat{\varphi}$ is $\ker\varphi = \langle P_3 + sQ_3\rangle$. With $\zeta = e(P_3, Q_3)$, which has order d, we get $e(K_g, Q_3) = \zeta^u$ and $e(P_3, K_g) = \zeta^{us}$. The codomain of $\hat{\varphi}$ is E_A up to isomorphism, and isomorphisms preserve these subgroups. The implementation maps the codomain to the model with coefficient A, using $x \mapsto -x$ when the computed coefficient is $-A$. □

Proposition 6 (fixed schedule)**.** *In the basis $(S,0),(T,0),(0,S),(0,T)$ of $E_C^2[2^E]$, the coordinates of $\ker\Phi$ are the columns of $\begin{pmatrix} c' & 0 \\ 0 & c' \\ -\lambda_S & 0 \\ 0 & -\lambda_T \end{pmatrix}$ modulo 2^E, which depend on sk only. Whether each $(2,2)$-step of the chain is a product step, a gluing, a Richelot step or a splitting is decided by linear conditions on these coordinates modulo powers of 2. So the schedule is the same for every ciphertext under a given key.*

Proof sketch. The kernel of the j-th step is 2^{E-j} times the image of $\ker\Phi$ under the first $j-1$ steps. On a product $E_1 \times E_2$, the step is a product step exactly when its 2-torsion kernel is a direct product $\langle(P,0),(0,Q)\rangle$, and a graph step when it is the graph of the identity of $E_C[2]$ (on $E_C \times E_C$) or of an isomorphism (after the split). Both are statements about the coordinates. On a Jacobian, the step is a splitting exactly when the codomain is a product, which happens at a fixed index determined by the same 2-adic data. Key generation imposes $\max(v_2(c' \mp \lambda_S)) = 2$ and the gluing test of line 15, which fixes the schedule as graph, product, glue, Richelot^{E-6}, split, product, graph. The self-test checks it once. □

Correctness error. By the propositions above, decapsulation of an honest ciphertext returns $H_K(s, ct)$ unless an intermediate formula meets a degenerate input. Examples are a Mumford divisor not in general position, a zero denominator in a Richelot or splitting formula, or a Miller line through the second argument. No branch handles these cases. The output is then wrong but computed in the same time, and the FO check rejects it. Heuristically each such event has probability $O(1/p)$ per step, giving $\delta = O(E/p) \approx 2^{-1330}$ at level 1. Key generation excludes the one structured degenerate case, a scalar $\equiv -1$ modulo the point order, which the x-only ladder with y-recovery cannot handle. The measured rates are in Table 5: no failure occurred in 3,635 decapsulations.

5 Parameters

The parameter sets are produced by `crypto:eigen_seta/examples/gen_params.rs`, which writes `crypto:eigen_seta/src/params_generated.rs`. They satisfy four constraints.

(P1) $d = 3^F \geq 2^{2\lambda}$, with $F = \lceil 2\lambda/\log_2 3\rceil$. Message recovery costs about $\sqrt{d}$ (§6.4).

(P2) $2^E > 28\,d^2 f_{\max}^2$ with $f_{\max} = 2f_{\text{lo}}$. Then $m = 2^E - 7d^2f^2 > \frac{3}{4}2^E$ for every admissible f, so the two-squares step has a large target. E is the smallest exponent that works.

(P3) $f_{\text{lo}} = 2^\kappa\lfloor p/2^E\rfloor + 1$. Then $fg \gtrsim 2^\kappa p$, and the norm equation $N(x) + pN(y) = fg$ has about 2^κ candidate values of $y \neq 0$. We take $\kappa = 12$ for the real sets and $\kappa = 8$ for the toys.

(P4) $p = 2^{E+1}3^F c - 1$ is prime with $p \equiv 3, 5, 6 \pmod 7$, and c is the smallest cofactor that achieves this.

Since $f \approx 2^{\kappa+1}cd$ and $2^E \approx 28d^2f^2$, we get $p \approx 2^{E+1}d \approx d^5 \cdot 2^{2\kappa+O(1)}c^2$. That is $\log_2 p \approx 10\lambda$. The factor d^5 is the price of decrypting with a single chain, whose torsion order must exceed the degree $7d^2f^2 + c^2$ of γ. Evaluating Φ from two half-chains, as in SQIsign 2D-style implementations [DMPR24], would let 2^E shrink to about $\sqrt{\deg \gamma}$. We have not implemented this.

Table 1: Parameter sets. $d = 3^F$; "f bits" is the bit length of the secret degree in the recorded keys; public key and ciphertext are four $\mathbb{F}_{p^2}$ elements each. Source: `crypto:eigen_seta/src/params_generated.rs`; f bits from `crypto:research/eigen_seta_rust_20261011/reports/run-20261011T1813Z-bench/records/bench_output.txt`.

set	λ	$\log_2 p$	E	F	$\log_2 d$	c	κ	f bits	pk = ct (B)
Toy16	16	216	174	21	33.3	158	8	51	216
Toy32	32	369	297	41	65.0	55	8	80	376
L1	128	1341	1076	162	256.8	144	12	278	1344
L3	192	1991	1594	243	385.1	1059	12	409	1992
L5	256	2630	2106	324	513.5	380	12	536	2632

The toy sets have no security; they exist so that every code path can be tested quickly.

6 Security analysis

This section is heuristic. We state the problems the scheme relies on, reduce key recovery to finding f, and estimate the cost of every attack we know of. Table 2 summarises the estimates. A cost written $2^x \cdot C$ means 2^x evaluations of a polynomial-time subroutine C, typically a $(2,2)$-chain in dimension 2 or an isogeny chain in dimension 8.

6.1 Problems

Assumption 1 (hidden-degree eigenbasis problem)**.** Given the public parameters and $pk = (E_A, P_A, Q_A)$, it is hard to find f, or any secret key that decapsulates honest ciphertexts for pk.

Assumption 2 (masked eigen-torsion problem)**.** Given pk and $ct = \mathsf{Enc}(pk, s)$ for uniform $s \in \mathbb{Z}/d$, it is hard to find s. This is the one-wayness of the deterministic PKE inside the KEM.

Lemma 7 (the degree is the key)**.** *Given f, a working secret key can be computed in polynomial time, up to a choice among the few representations of $m = 2^E - 7d^2f^2$ as a sum of two squares.*

Proof. $s_0 = fs_D$ is public once f is known. The pair (c, c') is one of the representations $c^2 + c'^2 = m$; key generation finds one with the same procedure an attacker would use, and the schedule condition and the signs leave at most a handful of candidates. Each candidate can be tested by decapsulating a self-made ciphertext. Neither α, ε nor λ_1, λ_2 is needed. □

So Assumption 1 is equivalent to the hardness of finding f. A consequence is that E_A and the eigen-structure must hide f; the next subsection shows what goes wrong when they do not.

6.2 The public-degree variant is insecure

The toy prototype of [Ano26] had a "walk" variant: E_A was the codomain of a secret isogeny ε of *public* degree 5^k, and $\tau = \varepsilon\omega\hat{\varepsilon}$. Its decryption key (c, c', s_0) is computed from $\Delta' = 4\deg\tau - (\mathrm{Tr}\,\tau)^2 = 7 \cdot 25^k$ and from nothing else, as the merged code shows (`crypto:research/eigen_seta_prototype_20261011/eigen_seta_toy.py`, lines 133–155 and 201–202). Since k is public, anyone can compute the key and decrypt. An exploratory run, not archived, decrypted

5 of 5 toy ciphertexts this way. Version 2 exists to close this hole: the degree of τ is $2f^2$ with f secret.

6.3 Key recovery

Exhaustive search over f. An attacker can enumerate candidates f, apply key generation's public filters, derive a candidate key by Lemma 7, and test it with one decapsulation. The filters accept a fraction ρ of the odd candidates in $[f_{\mathrm{lo}}, 2f_{\mathrm{lo}})$. We measured ρ empirically (`crypto:research/eigen_seta_rust_20261011/reports/run-20261011T1955Z-verify/records/f_filter_rate.txt`): $\log_2 \rho = -8.13$, -8.64 and -9.02 at L1, L3 and L5 (measured from 200, 100 and 100 accepted candidates, so about ± 0.1 to ± 0.14 binomial standard error). This leaves about $2^{267.8}$, $2^{398.6}$ and $2^{525.1}$ admissible degrees, an upper bound since the later rejections of key generation are not counted. The search costs that many chain evaluations classically and their square root with Grover's algorithm. We know of no test that rejects a wrong f more cheaply than a full chain. In the toy experiment of [Ano26, §6], 85 of 86 wrong but pairing-consistent guesses, including guesses agreeing with the truth on 3–13 low bits, ran the whole chain and failed only at the final split (`crypto:research/eigen_seta_prototype_20261011/results/`).

Recovering ε from a path. Let I be any left $\mathrm{End}(E_D)$-ideal connecting E_D to E_A. The isogeny ε corresponds to an equivalent ideal of norm f. There are about $\frac{15}{2\pi^2}X^2$ cyclic isogenies of degree at most X from E_D, spread over about $p/12$ curves, so on average about $9X^2/p$ of them land on a given curve. For $X = 2f_{\mathrm{lo}}$ this is about 2^{-782} at level 1. So ε is, with overwhelming probability, the unique shortest isogeny from E_D to E_A, and a reduced basis of I yields it, and f, in polynomial time. Key recovery therefore costs no more than finding *some* isogeny from E_D to E_A. That is the supersingular path or endomorphism-ring problem [EHL+18, Wes22]: $\tilde{O}(p^{1/2})$ classically [DG16] and $\tilde{O}(p^{1/4})$ quantumly [BJS14]. We also list $p^{1/3}$ as a conservative classical figure in view of recent heuristic algorithms with large memory [Ano26, §1.1]. Because $p \approx d^5$, all three are far beyond the target.

Meet in the middle on ε. If f is smooth, ε factors and a meet-in-the-middle search becomes possible. Even with f known and perfectly balanced, it costs at least $\sqrt{f} \geq \sqrt{f_{\mathrm{lo}}}$: $2^{138.5}$, $2^{204.1}$ and $2^{267.5}$ at the three levels, which is above 2^λ, and it applies only to the fraction of keys with smooth f.

Torsion-point attacks on the public key. The public key reveals ε on $E_D[2^E]$, with $2^{2E} \gg f$, up to the diagonal scalars λ_1, λ_2. The SIDH attacks [CD23, MMP+23, Rob23] need the degree to build the auxiliary isogeny of degree $2^E - f$ or $N^2 - f$. With f unknown they reduce to the exhaustive search above. Hiding the degree as a countermeasure to these attacks also appears in terSIDH and binSIDH [BF23], where the secret degree ranges over products of small prime powers; here it ranges over an interval. Knowing f would make λ_1, λ_2 irrelevant: τ acts on (P_A, Q_A) as $f \cdot \mathrm{diag}(\mu, \nu)$ whatever the scaling, and the attacker could simply decrypt.

Leakage through the public key. The Weil pairing $e(P_A, Q_A) = e(P_D, Q_D)^{\lambda_1\lambda_2 f}$ is uniform among primitive 2^E-th roots of unity, because $\lambda_1\lambda_2$ is a uniform unit. The eigenvalues of τ are $f\mu$ and $f\nu$, with μ, ν public, so an attacker knows the action of τ on $E_A[2^E]$ up to the unknown scalar $f \bmod 2^E$. We do not know how to use this without f. The decryption identity $\deg\gamma + c'^2 = 2^E$ involves f^2 through $\deg\tau$, so there is no smaller Kani instance that tests f modulo 2^j. This is the main structural assumption of the scheme, and it is where we would look first for an attack.

6.4 Message recovery

Mask guessing plus a torsion attack (binding). The ciphertext contains $S = a\varphi P_A$ and $T = b\varphi Q_A$, with φ of public degree d. The pairing $e(S,T) = e(P_A, Q_A)^{abd}$ reveals ab, but the ratio $r = b/a$ is uniform. If $r \bmod 2^n$ were known, the attacker would hold φ on $E_A[2^n]$ up to the common scalar a. Robert's attack in dimension 8 [Rob23] then recovers φ in polynomial time as soon as $2^{2n} > d$. With r odd, guessing it costs 2^{n-1} trials, where $n = \lfloor(\log_2 d)/2\rfloor + 1$. That is about 2^{128}, 2^{192} and 2^{256} dimension-8 chains classically, and the square root of that with Grover. FESTA's analysis [BMP23] rests on the same estimate. An attacker who knew τ could do better: the masks act diagonally on eigenvectors of $\theta = \varphi\tau\hat{\varphi}$, which is exactly what decapsulation exploits. But knowing τ means knowing f.

Claw finding on φ. Both ends of φ are public and its degree is d. A generic meet in the middle costs about $\sqrt{d}$ time and memory classically. Tani's algorithm costs $d^{1/3}$ quantum time with as much quantum memory, but under realistic cost models it was not better than classical van Oorschot–Wiener search [vOW99] for SIKE [JS19]. Our d is $2^{256.8}$, $2^{385.1}$ and $2^{513.5}$, which exceeds the degrees 2^{216}, 2^{305} and 2^{372} of SIKEp434, p610 and p751 [J$^+$20]. Those were assessed against the same attack for categories 1, 3 and 5. So claw finding is not the binding attack.

Eigenbases of known endomorphisms. The Castryck–Vercauteren attacks on FESTA and M-SIDH [CV23] need an endomorphism of E_A of small degree for which (P_A, Q_A) is an eigenbasis. The only such endomorphisms we know are those in $\mathbb{Z}[\tau]$, of degree at least $2f^2$, and they are not available without f.

6.5 Quantum security and NIST categories

NIST defines categories 1, 3 and 5 by the cost of key search on AES-128, -192 and -256 [Nat16]. Quantumly, that is Grover search over 2^{128}, 2^{192} or 2^{256} keys with an AES circuit as the oracle. The binding attack above has the same shape: a Grover search over about 2^λ candidates, but with an oracle (an isogeny chain in dimension 8 over a 1341- to 2630-bit field) that is far more expensive than an AES evaluation. Classically it is about 2^λ such chains against 2^λ AES evaluations. We therefore *conjecture* that L1, L3 and L5 meet categories 1, 3 and 5. Two caveats apply. The margin rests on the oracle cost alone: the search space matches the AES key space and is not larger. And any attack that guesses less than $\sqrt{d}$ of mask information would break the claim.

6.6 Chosen-ciphertext security and decryption failures

The KEM is $U^{\not\perp}[\mathsf{Enc}, G]$ [HHK17]. In the random-oracle model it is IND-CCA secure if Enc is one-way (Assumption 2) and has negligible correctness error δ; quantum random-oracle bounds for implicit rejection also exist [BHH$^+$19]. The adaptive key-recovery attacks that broke static SIDH keys [GPST16] send malformed ciphertexts and watch the result. Here every ciphertext that differs from $\mathsf{Enc}(pk, s)$ for the recovered s yields the pseudorandom $H_R(z, ct)$, in constant time (§7). Decapsulation must also never fail *observably* on malformed inputs. The robustness test (§8) feeds 48 adversarial ciphertexts, including singular and special curves, small-order points and random curves: none causes a panic, and none yields the honest key. Degenerate formula inputs do not give an attacker a failure oracle either. A crafted ciphertext is answered with $H_R(z, ct)$ whether or not an intermediate formula degenerates, because either way the re-encryption differs. Only the failure of an *honest* ciphertext would be visible, as a mismatch with the encapsulated key. That has probability heuristically $O(E/p)$ (§4), and it did not occur in 3,635 decapsulations. So decryption-failure attacks have nothing to work with.

Table 2: Estimated attack costs, as $\log_2$ of the number of basic operations. "chain" means one isogeny-chain evaluation. All figures are derived from the parameters, except ρ, which is measured. Source of ρ: `crypto:research/eigen_seta_rust_20261011/reports/run-20261011T1955Z-verify/records/f_filter_rate.txt`.

target	attack	L1	L3	L5
key	search over admissible f, classical (chains)	267.8	398.6	525.1
key	same with Grover (chains)	133.9	199.3	262.5
key	path to E_A then shortest ideal, classical $p^{1/2}$ / $p^{1/3}$	670 / 447	995 / 663	1315 / 876
key	path to E_A then shortest ideal, quantum $p^{1/4}$	335	498	657
key	meet in the middle on ε ($\sqrt{f}$)	138.5	204.1	267.5
message	mask guess + dim-8 attack, classical (chains)	128	192	256
message	same with Grover (chains)	64	96	128
message	claw on φ, classical $\sqrt{d}$	128.4	192.6	256.8
message	claw on φ, quantum $d^{1/3}$ (with $d^{1/3}$ qubits)	85.6	128.4	171.2
target λ (classical) / NIST category		128 / 1	192 / 3	256 / 5

6.7 Weak keys

- $\alpha \in \mathbb{Z}[\omega]$, i.e. $y = 0$. The Kani factor would then be an $\mathcal{O}_K$-ideal isogeny, and since $h(-7) = 1$ its codomain is E_D itself (or its conjugate). Key generation forces $(y_0, y_1) \neq (0, 0)$.
- $f \equiv g \pmod{d}$. The pairing test could not tell the two Kani factors apart, so key generation rejects these f via $2f \not\equiv 2^E$.
- Scalars $\equiv -1$ modulo the point order. These break y-recovery in the ladder; key generation rejects them.
- The schedule condition and the gluing test. These fix the 2-adic shape of $(c', \lambda_S, \lambda_T)$ for all keys. They are public by design, and their cost in key entropy is included in ρ.

7 Implementation and side channels

The implementation is the Rust crate `crypto:eigen_seta/` (about 4,300 lines plus tests). It is generic over the limb count, so one code base serves all parameter sets. Variable-time big integers (`num-bigint`) appear only in key generation and in public constants.

7.1 Arithmetic

- $\mathbb{F}_p$. Montgomery multiplication. For our primes $p + 1 \equiv 0 \bmod 2^{64Z}$, where Z is 16, 24 and 32 limbs out of 21, 32 and 42, so the reduction skips the zero limbs of $p + 1$ ("Montgomery-friendly" primes) and costs $N(N - Z)$ instead of N^2 word products.
- *Inversion.* Bernstein–Yang safegcd [BY19], generalising the structure of libsecp256k1's `modinv64` to any limb count, with the proven divstep bound. It is 40 to 140 times faster than Fermat inversion at L1–L5 (Table 4).
- $\mathbb{F}_{p^2}$. Karatsuba multiplication, inversion via the norm, and the constant-time square root of Adj and Rodríguez-Henríquez [ARH14].
- *Curves and isogenies.* x-only Montgomery ladders, including a three-point ladder, with Okeya–Sakurai y-recovery [OS01]; SIDH-style 3-isogeny chains with optimal strategies [CLN16]; Weierstrass arithmetic with constant-time selection between the

doubling and addition formulas; and the gluing, Richelot and splitting maps of [CD23] in Mumford coordinates, with batched inversions. The Weil pairing uses a projective Miller loop over the public bits of 3^F. Discrete logarithms in μ_{3^F} are computed digit by digit with constant-time comparisons.

7.2 Constant-time decapsulation

Decapsulation is split into two functions so that each can be checked separately.

- `decaps_prepare` depends only on (pk, ct). It parses both, lifts S and T, and computes both canonical bases. It runs in variable time.
- `decaps_core` contains everything secret: the Kani chain with its fixed schedule, the identification over the 8 sign and order choices, the pairings and DLPs, the 3-isogeny chain, the re-encryption, and the selection.

In the core, no branch and no memory index depends on secret data, and all loop counts are public. Selections use `subtle`'s constant-time primitives, and every mask passes through `core::hint::black_box`. Scalars have fixed widths. Secret scalars and secret byte buffers are wiped on drop. The only loop whose trip count depends on data, the canonical-basis search, runs on public curves and is capped. A crafted public key or ciphertext makes it return a refusal rather than spin.

A compiler-introduced leak. The masks in safegcd's final normalisation (add m if negative, negate if requested) are computed arithmetically, as `r + (m & mask)`. LLVM compiled the add into a conditional that skipped the work when the mask was zero, so the instruction count depended on the sign of a secret-dependent intermediate. That leak affected every field inversion. The source had no branch. The instruction-count probe of §8 found it, and annotating the probe at line level located it. Passing the masks through `black_box` fixed it. This is a reminder that constant-time code must be checked on the binary.

7.3 Residual issues

- Key generation is variable time: rejection sampling, Cornacchia and primality tests. It must run where no attacker can time it.
- Allocation: the core allocates vectors whose sizes are fixed by the parameter set. glibc's allocator then executes a number of instructions that depends on the heap state left by earlier variable-time work, though not on secrets. A production implementation should use fixed buffers.
- Points and field elements on the stack are not wiped, only scalars and byte buffers.
- The memory-address side of constant time is argued, not measured: there are no secret-indexed tables. Power, EM and fault attacks are out of scope. The FO check catches faults that corrupt s, but not faults that skip the check.
- Constant time is verified for one compiler (rustc 1.97, release profile) and one target (x86-64). Other builds need the probe re-run.

7.4 Tests

The crate's tests compare the field arithmetic with a big-integer reference at 1, 2, 9, 21 and 42 limbs, the chain with direct endomorphism evaluation, and the pairing with bilinearity. They also run round trips at Toy16 and Toy32. Known-answer tests pin SHA3-256 digests of the public key,

ciphertext and shared secret produced from fixed seeds at Toy16, Toy32 and L1. The robustness tests feed decapsulation 48 adversarial ciphertexts (§6.6) and encapsulation 14 adversarial public keys; every call terminates without a panic. A regression test covers a key-generation hang that we fixed: Tonelli–Shanks accepted $z = 0$ as a non-residue.

8 Measurements

All measurements were taken on one host: an Intel Xeon at 2.80 GHz (shared cloud VM, 4 logical cores), using rustc 1.97.0 in release mode. Two runs are used. The benchmark run `run-20261011T1813Z-bench` used the code at commit `19c641d12`. The verification run `run-20261011T1955Z-verify` used commit `d504af28e`, which adds the key-generation checks and the buffer wiping. The later commit `72051ac67` only caps the public canonical-basis search; the known-answer tests confirm that it changes no output. Their records are under `crypto:research/eigen_seta_rust_20261011/reports/`.

8.1 Sizes and speed

Table 3: Wall time per operation on one thread, as the range over trials (trial count in brackets). Every decapsulation returned the encapsulated key. Record: `run-20261011T1813Z-bench/records/bench_output.txt`.

set	pk = ct (B)	keygen (s)	encaps (s)	decaps (s)
Toy16	216	0.38 [1]	0.005–0.008 [5]	0.12–0.19 [5]
Toy32	376	2.3 [1]	0.009–0.012 [3]	0.45–0.59 [3]
L1	1344	32.3 [1]	0.24–0.26 [3]	13.4–15.0 [3]
L3	1992	118.6 [1]	1.03 [2]	51.8–53.9 [2]
L5	2632	271.8 [1]	2.17–2.20 [2]	120.2–124.0 [2]

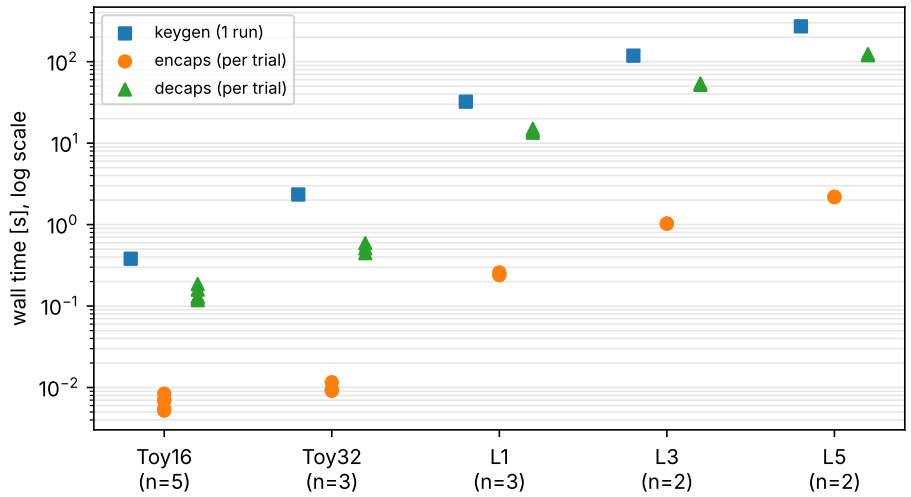


Figure 1: Wall time per operation, every trial shown, log scale. Measured; record as in Table 3.

Encapsulation is fast because it is a 3-isogeny chain of length F. Decapsulation is dominated by the $E - 6$ Richelot steps in Mumford coordinates. Each step inverts a polynomial modulo a quartic for every point it moves; from the formulas, we estimate a few hundred $\mathbb{F}_{p^2}$ multiplications

per point and step. Key generation is a single sample, and its time varies widely with the number of rejected candidates.

Table 4: $\mathbb{F}_p$ arithmetic in ns per operation, in the same binary and run. Multiplication is the minimum over 15 repetitions; inversion is the mean of 200. Record: `run-20261011T1813Z-bench/records/bench_output.txt`.

set	limbs	mul	inversion, Fermat	inversion, safegcd
Toy16	4	54	18,403	4,741
Toy32	6	72	67,657	10,316
L1	21	611	1,426,022	35,608
L3	32	1,516	6,708,390	109,506
L5	42	2,449	13,453,052	96,327

No pre-change binary was kept as a baseline for the KEM-level timings, so we claim no KEM-level speedup from the field optimisations.

8.2 Correctness

Table 5: Decapsulation outcomes for honest ciphertexts with fresh OsRng keys and ciphertexts. Records: `run-20261011T1955Z-verify/records/failrate/`, plus the benchmark run.

set	trials	failures	record
Toy16	3,005	0	verification run (3,000), benchmark run (5)
Toy32	603	0	verification run (600), benchmark run (3)
L1	23	0	verification run (20), benchmark run (3)
L3	2	0	benchmark run
L5	2	0	benchmark run

Zero failures in 3,005 trials bounds the failure rate at Toy16 only by about 10^{-3} at 95% confidence (rule of three). The argument that the rate is negligible is the heuristic $O(E/p)$ of §4; the trials show that no systematic failure mode exists.

Beyond these, the known-answer tests and the robustness tests pass: 48 adversarial ciphertexts to decapsulation and 14 adversarial public keys to encapsulation, with no panic and no hang (`crypto:eigen_seta/tests/`).

8.3 Constant-time evidence

Callgrind counted the instructions executed by `decaps_core` alone, with keys and ciphertexts generated from fixed seeds (`crypto:eigen_seta/examples/ct_probe.rs`). The count splits into glibc's allocator and `mem*` routines, whose work depends on heap layout but not on secrets (§7), and everything else.

A statistical timing test in the style of dudect [RBV17] (`crypto:eigen_seta/examples/dudect_decaps.rs`) used one Toy16 key and interleaved two classes in random order: a fixed ciphertext and a pool of 16 fresh ones, half of them FO-rejected. With 1,000 samples per class, Welch's statistic was +1.17 on the raw samples and at most 2.36 in absolute value after cropping at the 50th, 75th and 90th percentiles. The usual threshold for a detected leak is $|t| > 4.5$. This test is weaker than the instruction counts: it uses one key, wall-clock time on a shared VM, and compares ciphertext classes rather than keys. Record: `run-20261011T1955Z-verify/records/dudect_output.txt`.

Table 6: Instructions executed in the secret phase of decapsulation, excluding glibc's allocator. "Foreign" ciphertexts are well formed but made for another key, so the FO check rejects them. Records: `run-20261011T1955Z-verify/records/toyprobe/ctprobe_summary.txt` and `.../l1probe/ctprobe_summary.txt`.

set	keys	ciphertexts per key	non-allocator instructions
Toy16	4	2 valid, 1 foreign	1,116,209,216 in all 12 runs
L1	2	key 1: 2 valid, 1 foreign; key 2: 1 valid	120,996,429,610 in all 4 runs

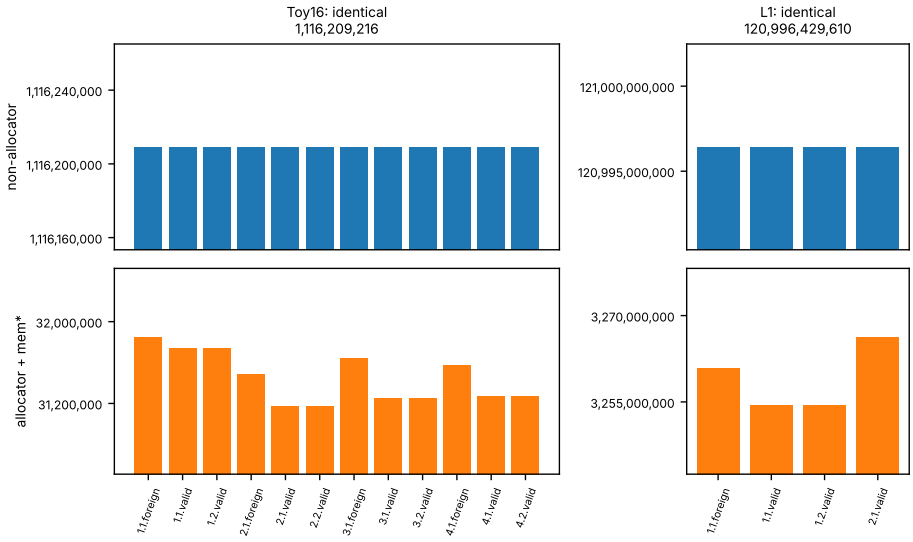


Figure 2: Instruction counts of `decaps_core` per run, at Toy16 (12 runs) and L1 (4 runs). Top: everything except glibc's allocator. Bottom: glibc's allocator and `mem*`. Measured; records as in Table 6.

9 Open problems

1. *Cryptanalysis of the hidden degree.* The scheme stands or falls with the claim that the public key gives no handle on f beyond exhaustive search (§6.3). In particular, the attacker knows the action of τ on $E_A[2^E]$ up to the scalar $f \bmod 2^E$. Independent analysis of this point is the most important next step.

2. *Mask entropy.* The binding attack guesses b/a modulo about $\sqrt{d}$. Any way to use fewer torsion bits, for example a dimension-4 variant of the torsion attack, or a lollipop built from another known endomorphism, would lower the security level directly.

3. *Smaller primes.* Decrypting with two half-chains would shrink 2^E from about $7d^2f^2$ to its square root and cut p from about d^5 to about d^3, by the derivation of §5 with $2^{2E} > \deg\gamma$.

4. *Speed.* Theta-coordinate $(2,2)$-chains [DMPR24] instead of Mumford coordinates, stack-only buffers, and SIMD field arithmetic.

5. *Constant-time key generation*, in particular a constant-time norm-equation solver.

6. *Compression* of public keys and ciphertexts, by encoding the points relative to canonical bases as in compressed SIDH.

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